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The flow of granular solids through orifices

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Abstract—The authors have compared the results of their own investigations on the flow of granular solids (a.o. a number of seeds) with results published by others. The flow of all the seeds tested through the circular orifices used in the experiments, is represented by the equation

$$W = 35\,\rho_B\sqrt{g}\,(D_0 - 1{\cdot}4\,d)^{2{\cdot}5}\ \text{(g/min)}$$

Résumé—Les auteurs comparent les résultats de leurs propres recherches concernant l'écoulement de matériaux granulaires (e.a. un nombre de semences) aux résultats publiés par d'autres auteurs. Le courant de toutes les semences examinées par les auteurs à travers les orifices circulaires employées dans la recherche est décrit au moyen de la formule :

$$W = 35\,\rho_B\sqrt{g}\,(D_0 - 1{\cdot}4\,d)^{2{\cdot}5}\ \text{(g/min)}$$

Zusammenfassung—Die Verfasser vergleichen die Ergebnisse eigener Versuche über das Ausströmen von Schüttgut (u.a. einige Samen) mit einer Anzahl von anderer Seite veröffentlichten Befunden. Der Durchfluss aller von den Verfassern untersuchten Samen durch kreisförmige Öffnungen konnte durch die Gleichung:

$$W = 35\,\rho_B\sqrt{g}\,(D_0 - 1{\cdot}4\,d)^{2{\cdot}5}\ \text{(g/min)}$$

erfasst werden.

1. INTRODUCTION

RECENTLY FOWLER and GLASTONBURY [1] published in this journal results of investigations on the flow of granular solids through orifices. In our laboratory also observations have been made on this subject, as part of a more comprehensive study on the flow of solids in general. It seems desirable to record our results in this article in order to compare them with those of other workers and to supplement their findings. FOWLER and GLASTONBURY [1] have published a comprehensive review of available literature on the subject, so that we will give only a survey of equations inferred by various other investigators. For comparison, the equations have been converted into the same system of units. Symbols and units are explained at the end of the paper. NEWTON *et al.* [2] found

$$W = 213\,D_0^{2{\cdot}96}\,H^{0{\cdot}04}\ \text{g/min}$$

FRANKLIN and JOHANSON [3] published for the flow from cylinders with flat bottoms, provided with orifices, the following equation:

$$W = \frac{1847\,\rho_s\,D_0^{2{\cdot}93}}{(6{\cdot}288\,\mu + 23{\cdot}16)(0{\cdot}394d + 1{\cdot}889) - 44{\cdot}90}\ \text{g/min}$$

BROWN and RICHARDS [4] also studied the influence of the shape of the orifice in the flat bottom of a cylinder. As a general equation they found for a material with $\rho_s = 2{\cdot}5$ g/cm^3

$$W = 4870\,A\,D_h^{0{\cdot}5}\,\psi\ \text{g/min}$$

and for circular orifices

$$W = 3730\,D_0^{2{\cdot}5}\,\psi\ \text{g/min}$$

Finally FOWLER and GLASTONBURY [1] found the equation

$$W = 14{\cdot}16\,\rho_B\,A\,\sqrt{2g\,D_h}\left[\frac{D_h}{d_s}\right]^{0{\cdot}185}\ \text{g/min}$$

We will return on these results further in this article.

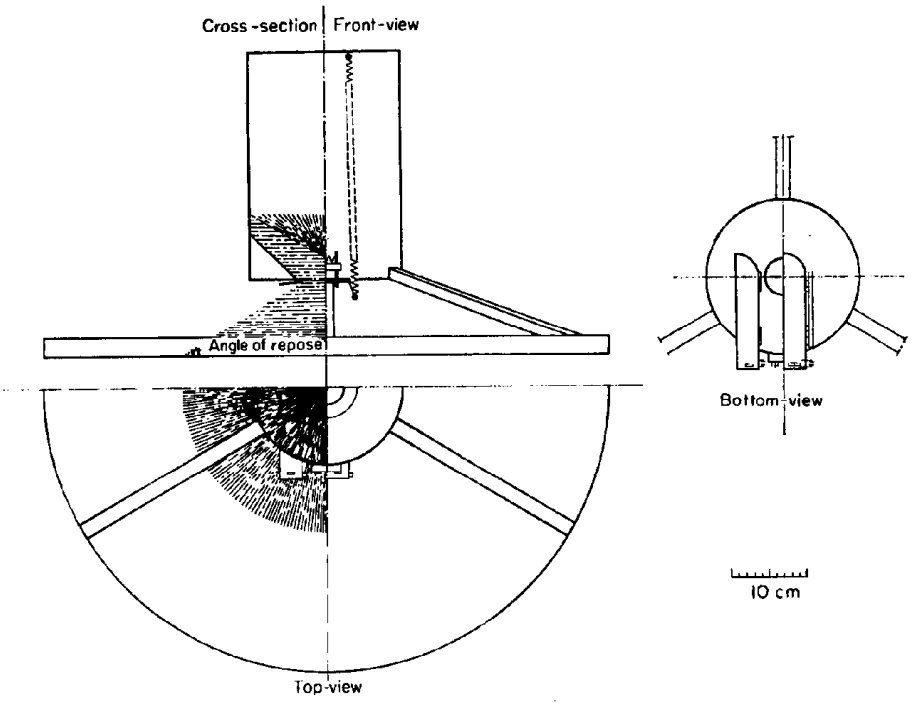


FIG. 2. Sketch of apparatus used for the determination of the angle of repose.

2. EXPERIMENTAL PROCEDURE AND MATERIALS

2.1 *Apparatus and Materials*

For an investigation of the flow of various materials use was made of three copper cylinders of 30 cm height and with internal diameters of respectively 5, 10 and 15 cm. In the flat bottoms of these cylinders were round openings which could be closed by means of valves. In these openings fitted copper disks with orifices differing in size and shape. In the cylinder with $D = 5{\cdot}0$ cm circular orifices with diameters of 0·20, 0·50, 0·75, 1·00, 1·25 and 1·50 cm could be fitted; in the cylinder with $D = 10{\cdot}0$ cm circular orifices with diameters of 0·50, 0·75, 1·00, 1·25, 1·50, 2·00, 2·50 and 3·00 cm. The largest cylinder ($D = 15{\cdot}0$ cm) could be provided with circular, square, rectangular and triangular orifices of dimensions as mentioned in Table 1.

The materials used in preliminary tests were sand-fractions and various seeds, viz. linseed, spinach (Juliana), watercress, rapeseed, kale, swede and turnip. Fig. 1 gives an impression of the shape and size of the seeds. As we could not predict exactly which properties of the materials influence their flow, we determined a number of characteristic properties, viz. the bulk density of packing, the true specific density, the void fraction (calculated from $100\,[(\rho_s - \rho_B)/\rho_s]$ per cent), the angle of repose, the number of particles per gramme and the particle size distributions of the seeds (by

Table 1
Details of orifices used in flow experiments

Circular orifices		*Square orifices* edge : a		*Rectangular orifices* side : a, 2a		*Triangular orifices* side : a, a, a,	
D_0 (cm)	A (cm²)	a (cm)	A (cm²)	a (cm)	A (cm²)	a (cm)	A (cm²)
0·26	0·056	0·22	0·049	0·16	0·055	0·38	0·064
0·50	0·20	0·49	0·24	0·35	0·24	0·74	0·24
1·03	0·84	0·89	0·80	0·63	0·80	1·46	0·92
1·50	1·76	1·35	1·84	0·99	1·96	2·06	1·84
2·03	3·24	1·77	3·16	1·25	3·16	2·79	3·36
2·50	4·96	2·22	4·96	1·58	5·00	3·52	5·36
3·02	7·20	2·65	7·04	1·90	7·28	4·02	7·00

screening with a perforated plate with circular openings and movement in a horizontal plane), from which the average screen size was calculated graphically. For the coarser materials the determination of the average screen size is less accurate than for the finer ones. For the determination of these properties current techniques were used. It should be mentioned only that the determination of the angle of repose at first presented many difficulties. The best results ultimately were obtained with the apparatus pictured in Fig. 2.

As we expected that also the shape of the particles influences the flow through orifices an attempt was made to express that shape into a "shape-factor." For this purpose the specific surface of the various materials was calculated from measurements of the resistance to the viscous flow of air through the materials according to Carman–Kozeny. Assuming that the particles were spherical, their "spherical diameter," d_s, was calculated from the specific surface and the void fraction. We defined the shape factor λ as the ratio of the average screen size of the particles to their "spherical diameter," i.e.

$$\lambda = \frac{d_s}{d} \quad (\lambda \leqslant 1)$$

The properties determined of the materials are presented in Table 2.

Table 2. Physical properties of the materials used in the flow experiments

Materials	*True density* ρ_s (g/cm³)	*Bulk density* ρ_B (g/cm³)	*Void fraction* (%)	*Average screen size d* (cm)	*Spherical diameter* d_s (cm)	*Shape factor* λ	*Angle of of repose* ϕ_s (degrees)	*Number of particles per* 10 g *N*
Sand	2·62	1·50	42·7	0·045	0·038	0·85	31·7	—
Linseed	1·16	0·69	40·5	0·25	0·12	0·48	30·9	1780
Spinach	1·19	0·57	52·1	0·30	0·16	0·54	31·3	1160
Watercress	1·22	0·71	42·6	0·16	0·11	0·71	30·4	3740
Rapeseed	1·12	0·67	40·2	0·17	0·15	0·91	28·1	4520
Kale	1·13	0·70	38·0	0·18	0·15	0·82	25·7	3250
Swede	1·09	0·66	39·4	0·18	0·14	0·75	25·6	3350
Turnip	1·12	0·68	39·3	0·17	0·13	0·79	25·8	4290

Table 3. Repeatability of the measurements

Material	*Orifice*	*Observed* (sec)	(g)	*W* (g/min)	*Average W* (g/min)	*Deviation* (%)
Sand 0·021 –0·030 cm	circular D_0 = 0·26 cm	240·2 240·0 239·6	112 111 111	28·0 27·8 27·7	27·8 ± 0·2	± 0·7
Watercress	square A = 7·04 cm²	17·8 16·8 16·8	2768 2620 2639	9380 9357 9425	9370 ± 50	± 0·5
Spinach	circular D_0 = 2·5 cm	37·0 38·3 37·0	2327 2436 2329	3773 3816 3777	3790 ± 25	± 0·7
Linseed	circular D_0 = 2·0 cm	61·2 60·7 60·4	2469 2452 2402	2421 2424 2386	2410 ± 25	± 1·0

The flow-rates W in g/min were determined three times by catching material during a definite time, after the flow had become stationary, and weighing the quantity caught. The examples given in Table 3 may be demonstrative for the repeatability of the results.

2.2 *Experimental results*

From Table 4 it appears that the head, within the range investigated (H = 10–30 cm), has no influence on the flow-rate. Also the diameter of the cylinder does not have an influence on the flow-rate; this can be seen from Table 5.

Tests with sand fractions showed (Fig. 3) that the particle size has only a slight influence on the flow, which influence decreases as D_0/d increases. When D_0/d amounts to about 20 or more the influence is negligible. The results mentioned

Table 4. Influence of head of packing

Material	D (cm)	D_0 (cm)	*Mass flow-rate* W (g/min)		
			H = 30 cm	H = 20 cm	H = 10 cm
Sand	5·0	0·25	34·7	34·3	34·8
0·0093–0·021 cm	5·0	0·50	248	252	245
	10·0	0·50	248	250	249
	10·0	1·00	1544	1548	1548
Sand	5·0	0·25	28·0	27·8	27·7
0·021–0·030 cm	5·0	0·50	217	220	218
	10·0	0·50	226	227	226
	10·0	0·75	646	647	650
	10·0	1·00	1497	1509	1524
	15·0	0·25	24·5	24·4	24·3
	15·0	0·50	283	283	283
	15·0	0·75	665	664	664
	15·0	1·00	1514	1515	1512
Sand	15·0	0·25	19·0	19·7	19·3
0·030–0·059 cm	15·0	0·50	203	204	203

Table 5. Influence of container diameter

Diameter of circular orifice D_0 (cm)	*Mass flow-rate* W g/min					
	Sand Container diameter			*Rapeseed Container diameter*		
	D = 5 cm	D = 10 cm	D = 15 cm	D = 5 cm	D = 10 cm	D = 15 cm
0·25	27·8	—	24·4	—	—	—
0·50	218	226	238	—	—	—
0·75	662	648	664	159	150	150
1·00	1492	1510	1508	399	403	405
1·25	2620	2622	2693	764	755	779
1·50	4323	4414	4260	1817	1876	1817
2·00	—	9072	9238	—	3082	3166
2·50	—	16580	16419	—	5844	5849
3·00	—	—	—	—	9721	9822

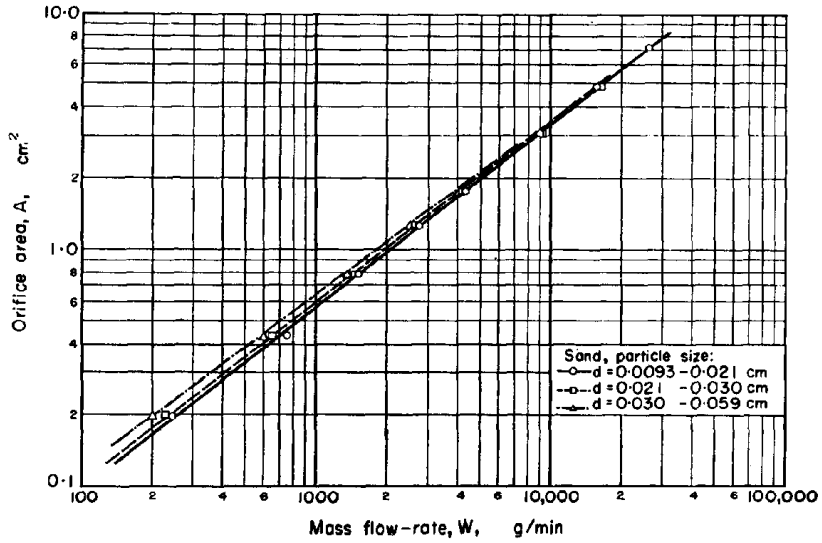


FIG. 8. Influence of orifice area on flow-rate of sand fractions.

above, are in agreement with those of earlier workers.

A large number of observations were made on the influence of the shape of the orifice. This influence was analogous with all materials investigated. Suffice therefore to give two examples, viz. of rapeseed and watercress (Figs. 4 and 5). It can be seen that with orifices of the same area the flow decreases in the order circle, square, rectangle and triangle. For mutual comparison of the materials circular orifices with diameters of at least six times the particle size were applied.

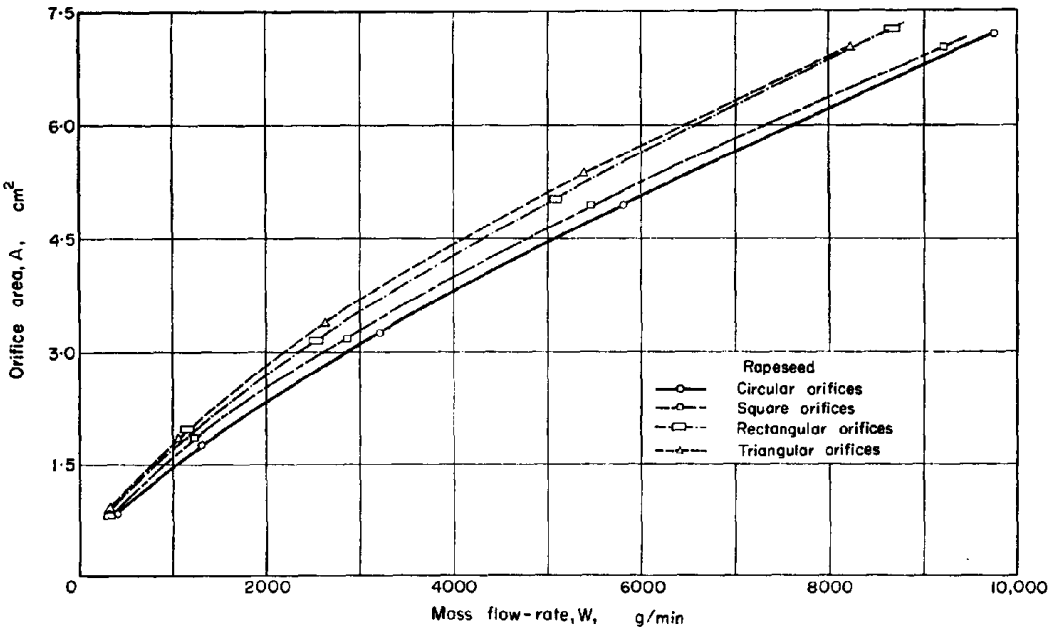


FIG. 4. Influence of orifice area on flow-rate of rapeseed.

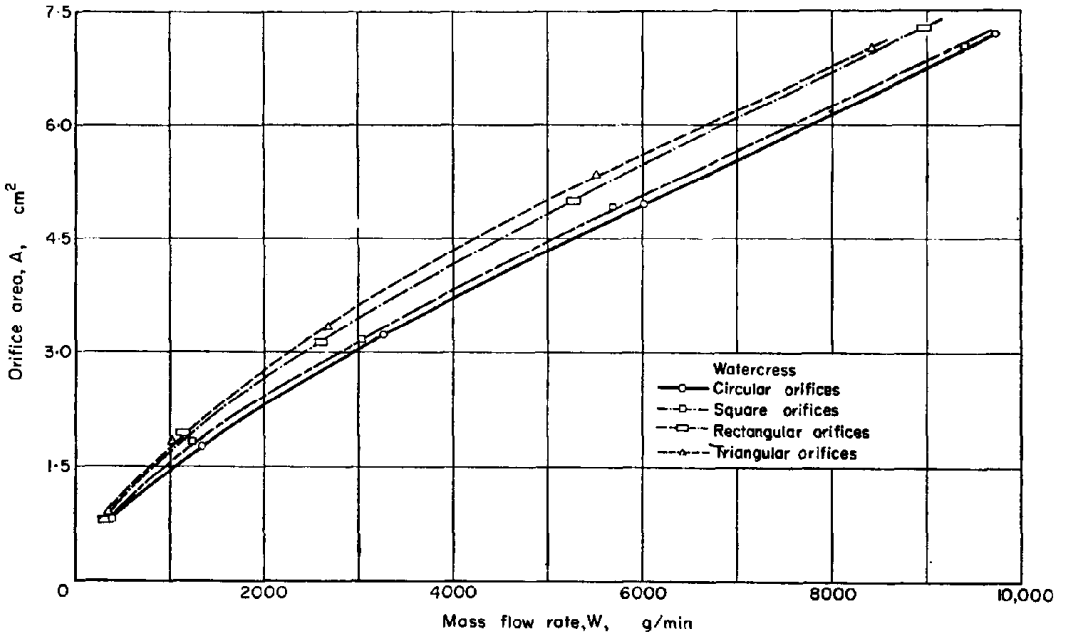


FIG. 5. Influence of orifice area on flow-rate of watercress.

The results of many observations have been presented in Fig. 6.

3. DISCUSSION

In agreement with earlier investigations we found that W plotted double logarithmically as a function of the area of the circular orifice A gives straight lines (Fig. 6). The same holds true for non-circular orifices.

A dimensional analysis suggests that W should

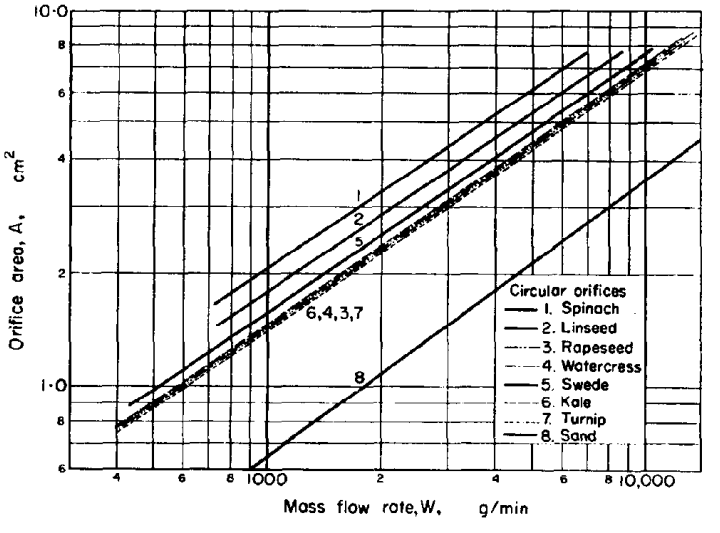


FIG. 6. Influence of orifice area on flow-rate of various materials.

be proportional to $\rho_B \sqrt{g}\, D_0^{2.5}$. However, from Fig. 6 we find exponents of D_0 greater than 2·5, particularly with larger d's, viz.:

sand	$W \div D_0^{2.77}$	rapeseed	$W \div D_0^{2.87}$
linseed	$W \div D_0^{3.09}$	kale	$W \div D_0^{2.85}$
spinach	$W \div D_0^{2.95}$	swede	$W \div D_0^{2.96}$
watercress	$W \div D_0^{2.90}$	turnip	$W \div D_0^{2.87}$

This might be explained by assuming that along the margin of the orifice a zone is useless or less fit for use for the flow. The size of that zone is proportional to d, so that the equation $W \div \rho_B \sqrt{g}\,(D_0 - k\,d)^{2.5}$ may be valid. In this equation k is a dimensionless constant.

By plotting D_0 against $W^{0.4}$ we found the following values for k:

sand	$k = 2.9$	rapeseed	$k = 1.4$
linseed	$k = 1.5$	kale	$k = 1.4$
spinach	$k = 1.4$	swede	$k = 1.5$
watercress	$k = 1.3$	turnip	$k = 1.4$

For all the seed samples examined we can therefore assume for k an average value of 1·4. The variations in k are probably related to the surface properties, but no correlation with the angle of repose could be established.

When we now introduce the factor ρ_B and $\sqrt{g}$ the following equations result:

sand	$W = 38.8\, \rho_B \sqrt{g}\,(D_0 - 2.9d)^{2.5}$
linseed	$W = 33.2\, \rho_B \sqrt{g}\,(D_0 - 1.4d)^{2.5}$
spinach	$W = 35.7\, \rho_B \sqrt{g}\,(D_0 - 1.4d)^{2.5}$
watercress	$W = 38.5\, \rho_B \sqrt{g}\,(D_0 - 1.4d)^{2.5}$
rapeseed	$W = 35.1\, \rho_B \sqrt{g}\,(D_0 - 1.4d)^{2.5}$
kale	$W = 36.3\, \rho_B \sqrt{g}\,(D_0 - 1.4d)^{2.5}$
swede	$W = 33.2\, \rho_B \sqrt{g}\,(D_0 - 1.4d)^{2.5}$
turnip	$W = 34.1\, \rho_B \sqrt{g}\,(D_0 - 1.4d)^{2.5}$

This shows that the coefficients in these equations are sufficiently constant to use for all the seed samples and for circular orifices the general formula

$$W = 35\, \rho_B \sqrt{g}\,(D_0 - 1.4d)^{2.5}$$

At first it seems remarkable that in the above equation the shape-factor does not appear at all. However, the influence of the shape finds expression more or less in the values of both ρ_B and d, so that in our opinion it is admissable to drop the shape-factor.

Calculations with this equation show for all the seeds within the scope of our investigations deviations of about 5 per cent. If we compare the coefficient in the general formula with the coefficients in the formulae given for the various materials separately it seems that the maximum deviation is 10 per cent. It should be considered, however, that the formulae given for the various materials do not describe the observations precisely, either, so the mutual deviation between the formulae is greater than the deviation between the general formula and the observations.

Using the effective hydraulic diameter D_h' so that $D_h' = D_h - 1.4d$ and the effective orifice area A', calculated from D_h', we find for the flow of rapeseed through orifices of various shapes the following equations:

circle	$W = 45.0\, \rho_B A' \sqrt{g D_h'}$
square	$W = 46.6\, \rho_B A' \sqrt{g D_h'}$
rectangle (1 : 2)	$W = 44.4\, \rho_B A' \sqrt{g D_h'}$
triangle (1 : 1 : 1)	$W = 45.7\, \rho_B A' \sqrt{g D_h'}$

It follows that the equation

$$W = 45\, \rho_B A' \sqrt{g D_h'}$$

may be applied for orifices of various shapes.

The slight influence of the particle size on the flow found for sand fractions (Fig. 3) is in agreement with the introduction of $(D_0 - k\,d)$ instead of D_0.

We cannot offer an explanation for the value of k for sand, which is somewhat different from 1·4; however, this value could not be calculated very accurately.

After having derived the general formula for seeds, mentioned above, we checked this formula with some other materials. For this purpose we used:

wheat	($d = 0.5$ cm, $\rho_B = 0.80$ g/cm³)
crystallized sugar	($d = 0.09$ cm, $\rho_B = 0.83$ g/cm³)
polystyrene globules	($d = 0.10$ cm, $\rho_B = 0.63$ g/cm³)
green peas	($d = 0.70$ cm, $\rho_B = 0.84$ g/cm³)
soybeans	($d = 0.66$ cm, $\rho_B = 0.75$ g/cm³)
lupin	($d = 0.65$ cm, $\rho_B = 0.70$ g/cm³)

From these data we calculated W with the equation

$$W = 35\,\rho_B\,\sqrt{g}\,(D_0 - 1{\cdot}4\,d)^{2{\cdot}5}$$

The following Table (6) shows the calculated and observed figures.

$$W = 35\,\rho_B\,\sqrt{g}\,(D_0 - 1{\cdot}4\,d)^{2{\cdot}5}$$

$$(\text{or } W = 45\,\rho_B\,A'\,\sqrt{g\,D_h'})$$

with that inferred by other workers it proves that our equation agrees best with the one according to FOWLER and GLASTONBURY [1], except for the

Table 6. Comparison between calculated and observed rate of flow for various granular materials

D_0 (cm)	*Wheat*		*Sugar*		*Polystyrene*		*Green peas*		*Soybeans*		*Lupin*	
	W_{calc}	W_{obs}	W_{calc}	W_{obs}	W_{calc}	W_{obs}	W_{calc}	W_{obs}	W_{calc}	W_{obs}	W_{calc}	W_{obs}
0·75	—	—	281	293	—	—	—	—	—	—	—	—
1·03	—	—	719	697	—	—	—	—	—	—	—	—
1·50	—	—	2010	2080	—	—	—	—	—	—	—	—
2·03	1790	1800	4610	4760	3400	3230	—	—	—	—	—	—
2·51	3820	3880	8050	8415	5980	5860	—	—	—	—	—	—
3·02	7210	6781	—	—	9830	9700	—	—	—	—	—	—
4·0	—	—	—	—	—	—	12590	14340	13540	13690	14220	12920
5·0	—	—	—	—	—	—	26180	29440	27060	27650	28870	26020
6·0	—	—	—	—	—	—	44890	51430	47000	47840	48520	44950

The differences between the experimental and the predicted flow rates in Table 6 amount to 5 per cent (average) and 12·5 per cent (maximum). From this we conclude that our formula can be used to make "calculated estimations" for the majority of materials with a bulk density ρ_B of about 0·7. If in any case the flow rate is to be known exactly, experiments will have to be made.

If we compare our equation

exponent. However, since the graphic representations cross each other in the field investigated (Fig. 7) the deviations within this field are not very great. In FOWLER's equation the influence of a useless zone along the margin of the orifice is expressed by multiplication with the dimensionless group $(D_h/d_s)^{0{\cdot}185}$ and in our equation by the term $1{\cdot}4\,d$, subtracted from D_h.

Attention is drawn to a discrepancy between

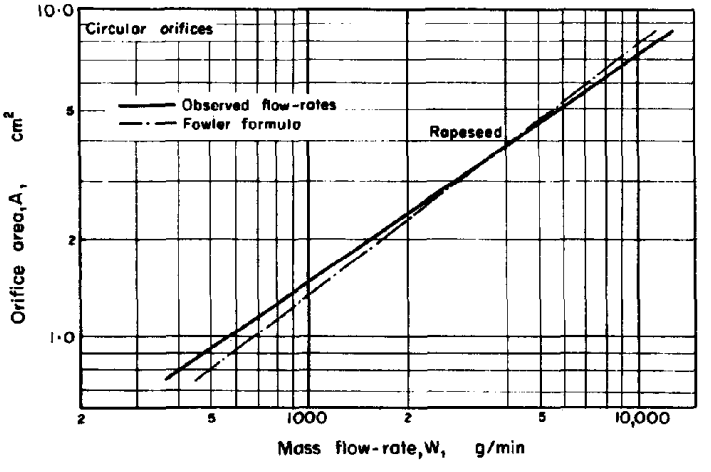


FIG. 7. Comparison of observed flow-rate of rapeseed with FOWLER's formula.

our shape factors λ (smaller than unity) and those according to FOWLER and GLASTONBURY [1] (above unity). In a following paper FOWLER and CHODZIESNER [5], however, use shape factors smaller than unity! All papers give the same definition of λ. For the formulae proposed above, however the discrepancy is immaterial as they do not contain λ.

NEWTON, *et al.* [2] have given an empirical formula for catalyst and found an exponent for D_0 of 2·96, which corresponds to the slope of the lines in our Fig. 6, e.g. for swede $W = 548\ \rho_B D_0^{2.96}$. NEWTON *et al.* found a very slight influence of the height, but this has not been confirmed, neither by us (Table 4) nor by others. Nor has an influence been found of the internal diameter of the container (Table 5).

According to FRANKLIN and JOHANSON [3] the exponent of D_0 should be 2·93. The correction for the particle diameter is incorporated in the denominator of their formula.

BROWN and RICHARDS' [4] correlation for sand with a specific gravity of 2·5 reads $W = 3730\ D_0^{2.5}\ \psi$ g/min (circular orifices). In the range of variables considered their reduced velocity ψ can be written

$$\psi = 0{\cdot}044 \left(\frac{D_0}{d}\right)^{0.60}$$

Consequently the flow rates measured are proportional to $D_0^{3.1}$, a particle diameter correction again being incorporated in the denominator.

ADDENDA

After the completion and revision of the present paper two recent articles came to our attention.

BROWN and RICHARDS [6] found that the local flow density is a function of the distance as measured from the edge of the effective aperture, this function being independent of the form of the aperture (circle or slot). This fact has been used implicitely in our correlations; we used a slightly different definition of the effective aperture.

BROWN and RICHARDS' [6] equation for the local flow density may be integrated over a circular area to give the total mass flow. If an average $\psi = 1{\cdot}00$ is used and the average ratio ρ_B/ρ_s is taken as 0·58 (as for our seeds), this formula for mass flow becomes.

$$W = 31{\cdot}7\ \rho_B \sqrt{g}\,(D - kd)^{2.5}\ \text{g/min}\ *$$

which is some 10 per cent below our formula.

LUK'YANOV *et al.* [7] gave a formula for the volumetric flow of spherical catalyst particles through a circular orifice. They corrected in a slightly different way for the inaccessible area at the edge of the aperture, their formula being

$$V = 5730\ \sqrt{R}\,(R^2 - 1{\cdot}9d\,R + 1{\cdot}66\,d^2)\ \text{cm}^3\text{/min}.$$

The numerical values according to this formula are about 7 per cent below ours.

*It should be noted that the symbol k in BROWN and RICHARDS' paper has the same meaning as kd in our formulae.

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Furthermore we wish to express our gratitude to Ir. A. HARMENS, who made valuable suggestions as to the method of correlation of our experimental data.

NOTATION

A = area of orifice	cm²
D = container diameter	cm
D_h = hydraulic diameter = 4 × A/perimeter of orifice	cm
D_0 = diameter of circular orifice	cm
d = average screen size of particles	cm
d_s = spherical diameter of particles	cm
g = gravitational constant	cm/sec²
H = head of packing above orifice	cm
k = an empirical coefficient	
λ = shape factor of particles $\lambda = d_s/d$	
N = number of particles/10 g	
ρ_B = bulk density of packing	g/cm³
ρ_s = true density of solid	g/cm³
μ = tangent to internal kinetic angle ϕ_i	
ϕ_s = angle of repose	
ψ = reduced velocity	
V = volume discharged/unit time	cm³/min
W = weight discharged/unit time	g/min

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